

Financial Impact Analysis

```
import pandas as pd

df = pd.read_csv('financial_impact.csv')
```

Variables

```
direct_losses = df['direct_loss_usd']
direct_loss_method = df['direct_loss_method']
recovery_cost = df['recovery_cost_usd']
total_loss = df['total_loss_usd']
inflation_adjusted = df['inflation_adjusted_usd']
insurance_payout = df['insurance_payout_usd']
regulatory_fine = df['regulatory_fine_usd']
total_loss_method = df['total_loss_method']
ransom_demanded = df['ransom_demanded_usd']
ransom_paid = df['ransom_paid_usd']
ransom_source = df['ransom_source']
```

Descriptive Analysis

Dataset Information

```
print(df.info())

<class 'pandas.DataFrame'>
RangeIndex: 778 entries, 0 to 777
Data columns (total 19 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   incident_id           778 non-null   str
 1   direct_loss_usd       778 non-null   float64
 2   direct_loss_method    778 non-null   str
 3   ransom_demanded_usd   206 non-null   float64
 4   ransom_paid_usd       86 non-null    float64
 5   ransom_source         86 non-null    str
 6   recovery_cost_usd     778 non-null   float64
 7   legal_fees_usd        778 non-null   float64
 8   regulatory_fine_usd   132 non-null   float64
 9   insurance_payout_usd  435 non-null   float64
10  total_loss_usd        778 non-null   float64
11  total_loss_method     778 non-null   str
12  total_loss_lower_bound 778 non-null   float64
13  total_loss_upper_bound 778 non-null   float64
14  inflation_adjusted_usd 778 non-null   float64
15  cpi_index_used        778 non-null   str
16  notes                 248 non-null   str
17  created_at            778 non-null   str
18  updated_at            778 non-null   str
dtypes: float64(11), str(8)
memory usage: 115.6 KB
None
```

- The information regarding the contents of the dataset.

```
# Direct Losses
print(direct_losses.describe())
```

```
count    7.780000e+02
mean     3.652294e+07
std      1.196191e+08
min      9.000000e+04
25%      2.834239e+06
50%      8.525071e+06
75%      2.609037e+07
max      2.302300e+09
Name: direct_loss_usd, dtype: float64
```

- Direct losses are right-skewed
- Median = ~8,525,071.00; —*Losses ranged from 90,000.00 to ~2,302,300,000.00* (2,302,210,000 difference)

```
# Recovery Costs
print(recovery_cost.describe())
```

```
count      7.780000e+02
mean       2.618556e+07
std        7.816093e+07
min        5.703142e+04
25%        2.055169e+06
50%        6.129803e+06
75%        1.840980e+07
max        1.238193e+09
Name: recovery_cost_usd, dtype: float64
```

- Recovery costs are right-skewed
- Median = ~6,129,803.00 – Range = 57,031.42 to ~\$1,238,193,000.00

```
# Total Loss and adjusted costs by inflation
print(total_loss.agg(["count", "median", "min", "max", "mean"]))
print("-----")
print(inflation_adjusted.agg(["count", "median", "min", "max", "mean"]))
```

```
count      7.780000e+02
median     1.656491e+07
min        1.737931e+05
max        3.451548e+09
mean       7.099600e+07
Name: total_loss_usd, dtype: float64
-----
count      7.780000e+02
median     1.790865e+07
min        1.737931e+05
max        3.451548e+09
mean       7.501910e+07
Name: inflation_adjusted_usd, dtype: float64
```

- **Total Losses**
 - Right-skewed distribution
 - Median = ~16,564,910.00 – Range = 173,793.10 - ~3,451,548,000.00 – ** AdjustedInflation ** – Right – skewed distribution – Median = 17,908,650.00
 - Range = ~173,793.10 – 3,451,548,000.00

```
# Insurance Payout
print(insurance_payout.describe())
```

```
count      4.350000e+02
mean       2.665044e+07
std        7.593903e+07
min        6.280104e+04
25%        2.277000e+06
50%        6.316200e+06
75%        1.844803e+07
max        1.054416e+09
Name: insurance_payout_usd, dtype: float64
```

```
# Ransom Demanded + Paid
print(ransom_demanded.agg(["count", "median", "min", "max", "mean"]))
print("----")
print(ransom_paid.agg(["count", "median", "min", "max", "mean"]))
```

```
count      2.060000e+02
median     1.979266e+06
min        5.000000e+04
max        7.500000e+07
mean       8.017672e+06
Name: ransom_demanded_usd, dtype: float64
----
count      8.600000e+01
median     1.129759e+06
min        1.610098e+04
max        5.188418e+07
mean       4.497464e+06
Name: ransom_paid_usd, dtype: float64
```

- 206 documented ransom demands
 - Median = ~1,979,266.00 – Range = 50,000.00 - ~75,000,000.00 – 86 documented paid ransoms – Median = 1,129,759.00

- Range = ~16,100.98 — 51,884,180.00

```
# Ransom Source
print(ransom_source.value_counts())
```

```
ransom_source
congressional testimony    19
blockchain analysis        19
FBI disclosure              15
company statement          13
insurance claim record     11
court filing                9
Name: count, dtype: int64
```

```
# Method for calculating total losses
```

```
print(total_loss_method.value_counts())
```

```
total_loss_method
estimated    369
disclosed    215
calculated   194
Name: count, dtype: int64
```

- The majority of total loss is estimated

```
print("\nMissing Values")
print("-----")

missing = df.isnull().sum()
missing_pct = (missing / len(df) * 100).round(2)

missing_df = pd.DataFrame({
    "Missing Count": missing,
    "Missing %": missing_pct
})

print(missing_df.sort_values("Missing %", ascending=False))
```

```
Missing Values
-----
Missing Count  Missing %
ransom_paid_usd      692    88.95
ransom_source        692    88.95
regulatory_fine_usd   646    83.03
ransom_demanded_usd   572    73.52
notes                530    68.12
insurance_payout_usd  343    44.09
total_loss_lower_bound    0     0.00
created_at            0     0.00
cpi_index_used          0     0.00
inflation_adjusted_usd    0     0.00
total_loss_upper_bound    0     0.00
incident_id            0     0.00
total_loss_method       0     0.00
total_loss_usd          0     0.00
direct_loss_usd         0     0.00
legal_fees_usd          0     0.00
recovery_cost_usd       0     0.00
direct_loss_method       0     0.00
updated_at             0     0.00
```

- The data describes all missing values in the dataset

